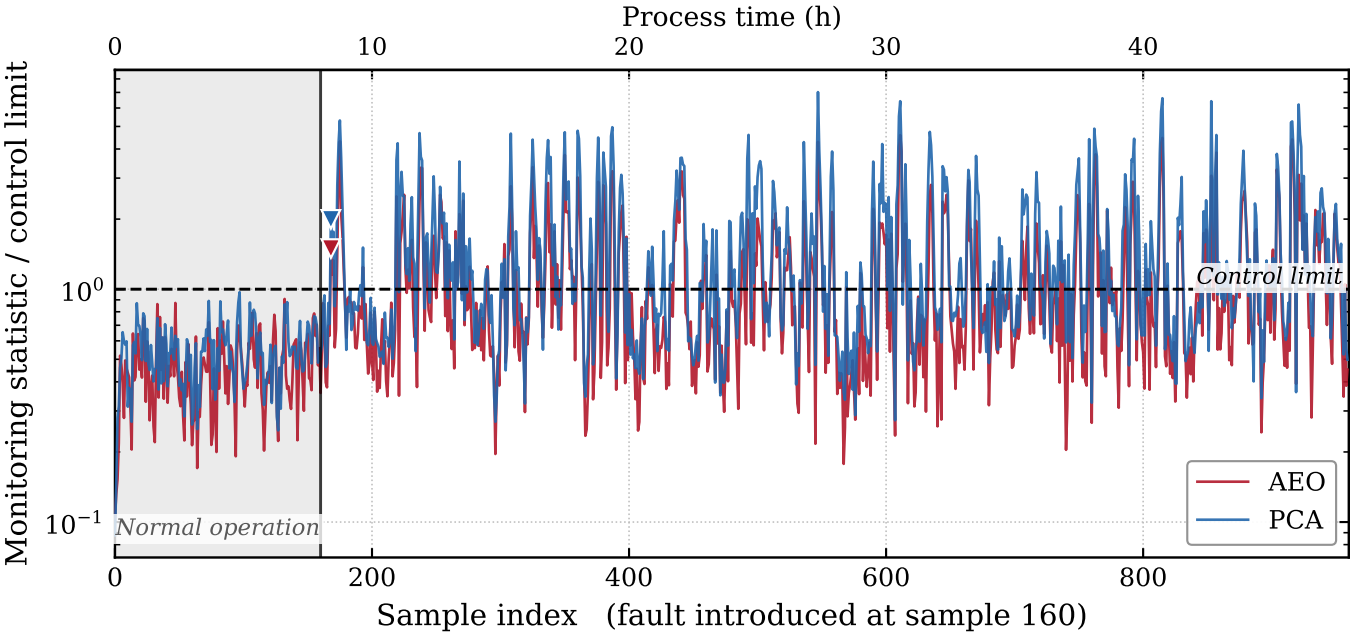
Fault Detection Performance for IDV(11)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	14.1 samples (42 min)	MDR 0.586	detected in 7/7 runs
PCA	delay	8.3 samples (25 min)	MDR 0.481	detected in 7/7 runs

PCA detects 5.9 samples (18 min) earlier; PCA missed-detection rate lower by 18%